Chess AI using Alpha-Beta Pruning and Min-max algorithm

Submitted by :
Shanu Yadav(MA23M013)
Gurmeet Singh(MA23M007)
Submitted to:
Dr. Ajin George Joseph

IIT Tirupati

April 30, 2025

What is Alpha-Beta Pruning?

- Alpha-Beta Pruning is an optimization technique for Minimax algorithm.
- It reduces the number of nodes that are evaluated.
- Branches that won't affect the result are ignored (pruned).
- This allows deeper search within the same computation time.

Project Overview and Challenges

- **Objective:** Build a Chess AI using Alpha-Beta Pruning .
- **Evaluation:** Material count, center control, mobility.
- **Challenges:**
  - Frequent Draws (1/2-1/2) due to weak endgame understanding.
  - Game getting stuck without enough search depth.
  - Difficulties in creating complete GIF animations.
- **Solution:**
  - Increased search depth to 4 and adjusted move limits.
  - Resulted in smoother gameplay and proper game completion.

Result and Demonstration for alpha-beta prunning

- Game usually ended in a **Draw (1/2 - 1/2)** or slow progress due to limited evaluation strength.
- Alpha-Beta Pruning improved search speed but could not guarantee a winner without stronger logic.
- **Final GIF demonstration:**

Video Link: `https://drive.google.com/drive/folders/1NAIiJpbC--Rmv8DlrlmftaiE7-cF2hUI?usp=drive_link`

What is Min-Max algorithm?

- The **Minimax algorithm** is a recursive strategy for two-player games, assuming both players play optimally.

Let s be a game state, and define the value function $V(s)$ as:

$$V(s) = \begin{cases} \text{Utility}(s) & \text{if } s \text{ is terminal,} \\ \max\limits_{a \in \text{Actions}(s)} V(\text{Result}(s, a)) & \text{if Max's turn,} \\ \min\limits_{a \in \text{Actions}(s)} V(\text{Result}(s, a)) & \text{if Min's turn.} \end{cases}$$

Where:

- $\text{Actions}(s)$ returns possible moves from s.
- $\text{Result}(s, a)$ returns the next state after applying move a.
- $\text{Utility}(s)$ is the evaluation of a terminal state.

Result and Demonstration for Min-Max algorithm

- Main functions used in my program are- search, minmax1, minmax2 and main program itself.
- The user needs to enter the move by adress. That is from where to where it is moving. For example 'g1h3', 'a2a4', 'd1d2' etc. Note that all moves should be valid otherwise the code will result into invalid move.
- After entering the address , user needs to press "Enter" and the move will be reflected into output. And then the next move will be by the algorithm.
- When user will lose or win, the code will ends with error that no moves are remaining.

Video Link: https://drive.google.com/drive/folders/14DkPpaxI2RhqxxIw00j-Buy042culFMG?usp=drive_linkE